LAKSHMI LAHARI SATTI

Sr. AI/ML ENGINEER | 5 YEARS

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PROFESSIONAL SUMMARY

Sr. AI/ML Engineer with **5 years of expertise** in architecting production-grade Generative AI systems, multi-Agent workflows, and MLOps pipelines. Proven track record at Scale AI and Haptik in delivering end-to-end autonomous solutions, including GraphRAG implementation and LLM fine-tuning (QLoRA), which reduced operational overhead by 40% and infrastructure costs by $120k annually. Expert in scaling distributed training on AWS/Azure and deploying optimized models via Kubernetes and TensorRT. Committed to Responsible AI practices, utilizing NIST frameworks and Explainable AI (XAI) to ensure regulatory compliance and algorithmic fairness across large-scale enterprise deployments. High-performance programmer in Python and C++ with a Master's from Arizona State University, focused on bridging the gap between cutting-edge research and scalable, metric-driven business impact.

TECHNICAL SKILLS

- **Agentic Workflows & Autonomy:** Multi-agent Systems, LangGraph, CrewAI, AutoGen, Agentic RAG, Tool-use (Function Calling), Human-in-the-Loop (HITL) Design, Autonomous Reasoning.
- **Generative AI & LLMs:** Large Language Models (LLMs), RAG Architectures, Fine-tuning (LoRA/QLoRA), LangChain, Vector Databases (Pinecone/Milvus), Prompt Engineering, Multi-modal Models.
- **MLOps & Production Engineering:** AWS SageMaker, Azure ML, Google Vertex AI, Docker, Kubernetes, MLflow, CI/CD for ML, Model Observability (Evidently AI, Arize), Drift Detection, TDD for ML.
- **Advanced Modeling & Optimization:** PyTorch, JAX, TensorFlow, Quantization (INT8/FP16), Model Pruning, Knowledge Distillation, TensorRT, ONNX, Hyperparameter Tuning (Optuna).
- **Data Architecture & Big Data:** Apache Spark, Databricks (Medallion Architecture), Snowflake, Kafka (Real-time Streaming), Feature Stores (Feast), ETL/ELT Pipelines, Distributed Training.
- **Edge AI & Hardware Acceleration:** TinyML, TensorFlow Lite, PyTorch Mobile, NVIDIA Jetson, CoreML, ARM Architecture Optimization, Latency & Power-Consumption Profiling.
- **Responsible AI & Governance:** NIST AI Risk Management Framework, Algorithmic Bias Mitigation, Explainable AI (XAI/SHAP), Differential Privacy, AI Auditing, Regulatory Compliance (EU AI Act/CCPA).
- **Core Programming & Math:** Python (Expert), SQL, C++ (High Performance), Java/Scala, Linear Algebra, Multivariable Calculus, Bayesian Statistics, Optimization Algorithms.

PROFESSIONAL EXPERIENCE

Sr. AI/ML Engineer | Scale AI ***July 2025 – Present | USA***

- Architected enterprise-grade multi-agent orchestrations using LangGraph and CrewAI, enabling autonomous decision-making workflows that reduced manual operational intervention by 40% across high-scale client production environments.
- Engineered high-performance RAG (Retrieval-Augmented Generation) pipelines utilizing GraphRAG and Milvus vector databases, improving retrieval precision by 35% and significantly reducing hallucination rates in complex domain-specific queries.
- Spearheaded model optimization strategies including QLoRA fine-tuning and Quantization (INT8/FP16), achieving a 50% reduction in inference latency and cutting cloud GPU infrastructure costs by $120,000 annually.
- Implemented robust MLOps frameworks using Kubeflow and Terraform to automate end-to-end model deployment, resulting in a 60% faster time-to-market for generative AI features within a Kubernetes ecosystem.
- Developed real-time model observability dashboards using Arize and Evidently AI to track data drift and prediction quality, ensuring 99.9% reliability for mission-critical AI applications in live production.
- Directed the integration of NIST AI Risk Management frameworks to ensure algorithmic fairness and compliance, mitigating bias in large-scale datasets and aligning with current global AI governance standards.
- Designed scalable data ingestion layers using Apache Spark and Kafka, processing over 5TB of unstructured data daily to power multi-modal model training for advanced computer vision applications.

AI/ML Engineer | Haptic AI ***Jan 2020 – Dec 2023 | India***

- Developed and deployed sophisticated predictive models utilizing PyTorch and Scikit-learn, which optimized lead scoring accuracy by 25% and directly contributed to a $1.5M increase in quarterly sales revenue.
- Built end-to-end computer vision pipelines for automated defect detection using YOLOv8 and TensorRT, increasing manufacturing inspection throughput by 45% while maintaining a 98% precision rate on edge devices.
- Optimized large-scale distributed training jobs on AWS SageMaker, reducing model convergence time by 30% through effective hyperparameter tuning with Optuna and strategic data sharding techniques across GPU clusters.
- Collaborated with cross-functional product teams to integrate NLP-based sentiment analysis tools into customer support interfaces, reducing average ticket resolution time by 20% through automated intent classification.
- Established a comprehensive DVC (Data Version Control) system to manage experimental reproducibility, allowing the engineering team to iterate on model versions 50% more efficiently without losing historical data integrity.
- Authored high-performance FastAPI wrappers for ML model serving, handling over 5,000 concurrent requests per second with sub-100ms latency, ensuring a seamless user experience for external API consumers.
- Revamped legacy ETL pipelines into modern Medallion Architecture using Databricks, enhancing data quality and reducing pipeline failure rates by 70% through automated validation and schema enforcement.

PROJECTS

Intelligent Document Classification and Sensitive Data Detection System

- Architected an automated PII/PHI (Personally Identifiable Information) detection system using SpaCy NER and Regex-based filters, achieving 99.8% accuracy in identifying sensitive data across 1M+ unstructured enterprise documents for CCPA and GDPR compliance.
- Engineered scalable Python preprocessing pipelines for raw text normalization and OCR integration, converting multi-format legacy files (PDF, Scans, Docx) into structured, high-quality inputs optimized for Transformer-based classification and embedding generation.
- Fine-tuned state-of-the-art DeBERTa-v3 and Longformer architectures on domain-specific datasets, delivering a 94% F1-score in categorizing long-form documents while outperforming baseline models in handling complex, multi-page legal and financial text sequences.
- Developed a high-performance Vector Search engine utilizing FAISS and Sentence-Transformers, enabling sub-second semantic retrieval across document collections to facilitate rapid clustering of sensitive information and duplicate detection within a containerized Docker environment.

EDUCATION

Master of Science in Computer Science **Jan 2024 – Dec 2025**

University of Arizona |Tucson, AZ